ANALYTICAL CUMULATIVE EXAMINATION

March 31, 1990

1) For the circuit shown in Figure 1, write an expression for E_0 in terms of Vr, Vrs and resistors R_1, R_2, R_3 and R_4.

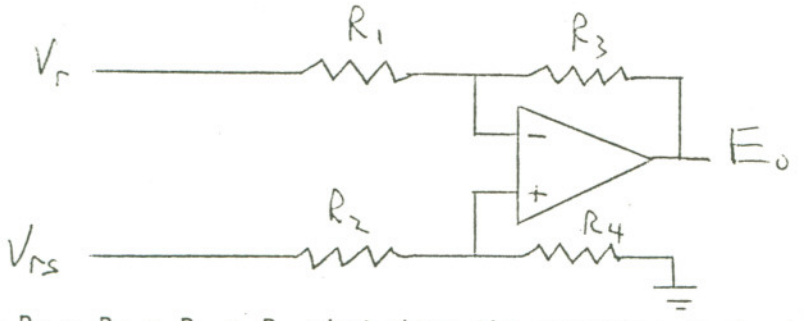


Figure 1

2) If $R_1 = R_2 = R_3 = R_4 = R$, what does the equation derived in (1), reduce to?

3) Shown below, Figure 2, is a sample and hold (SH) amplifier. When the control line is "high", the amplifier is in the "hold" mode; when the control line is "low", the amplifier is in the "sample" mode.

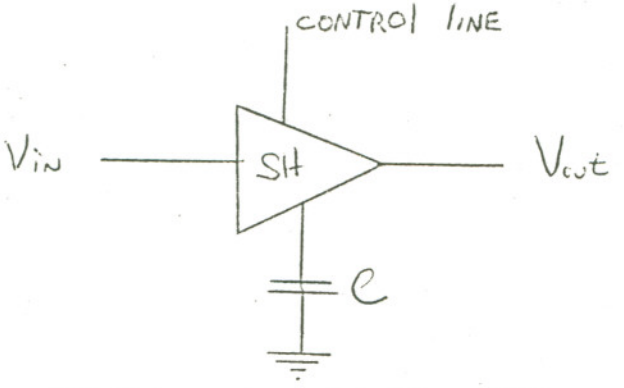
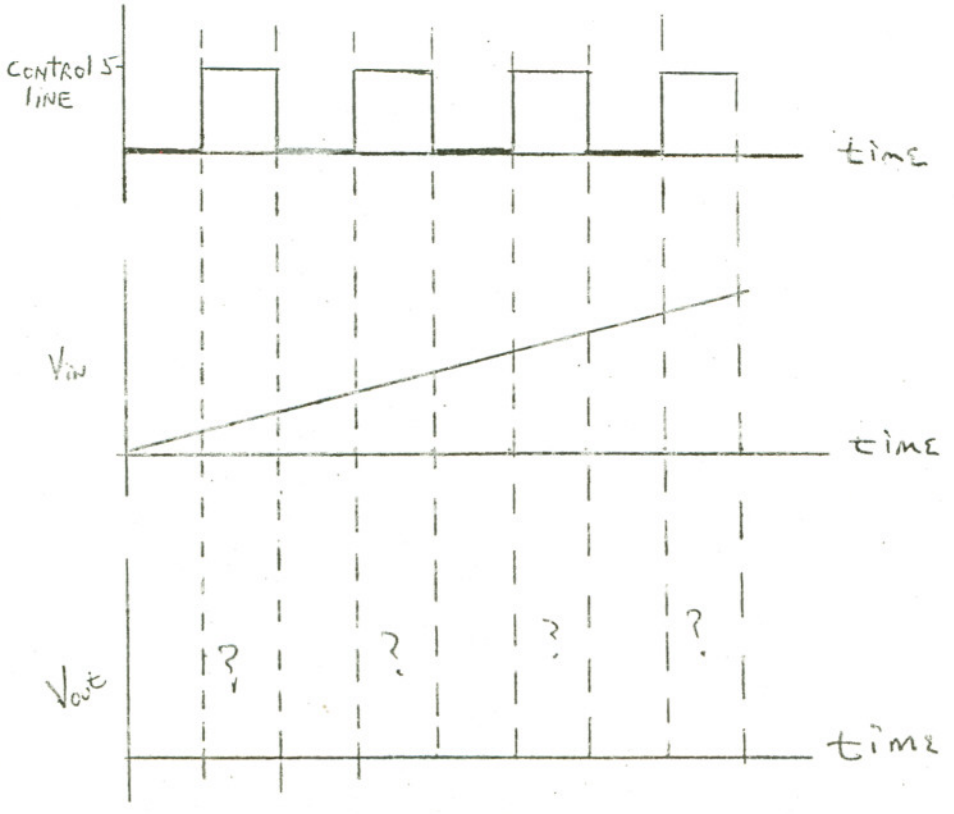


Figure 2

Trace the output, V_{out}, of the sample and hold amplifier given the following timing diagrams:



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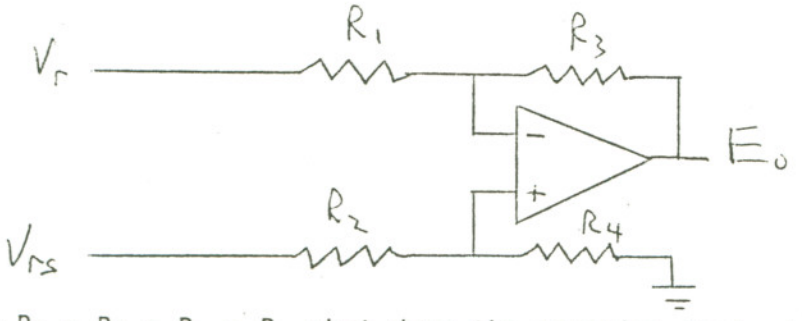

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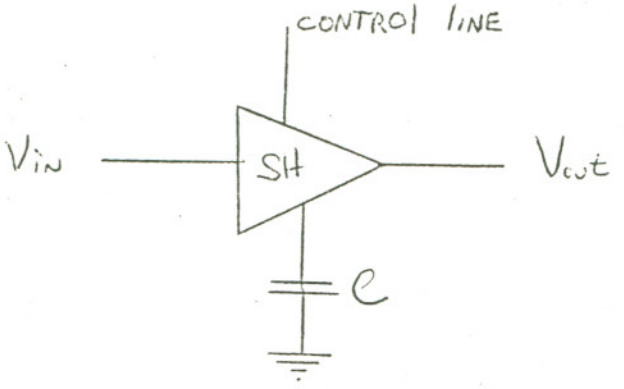

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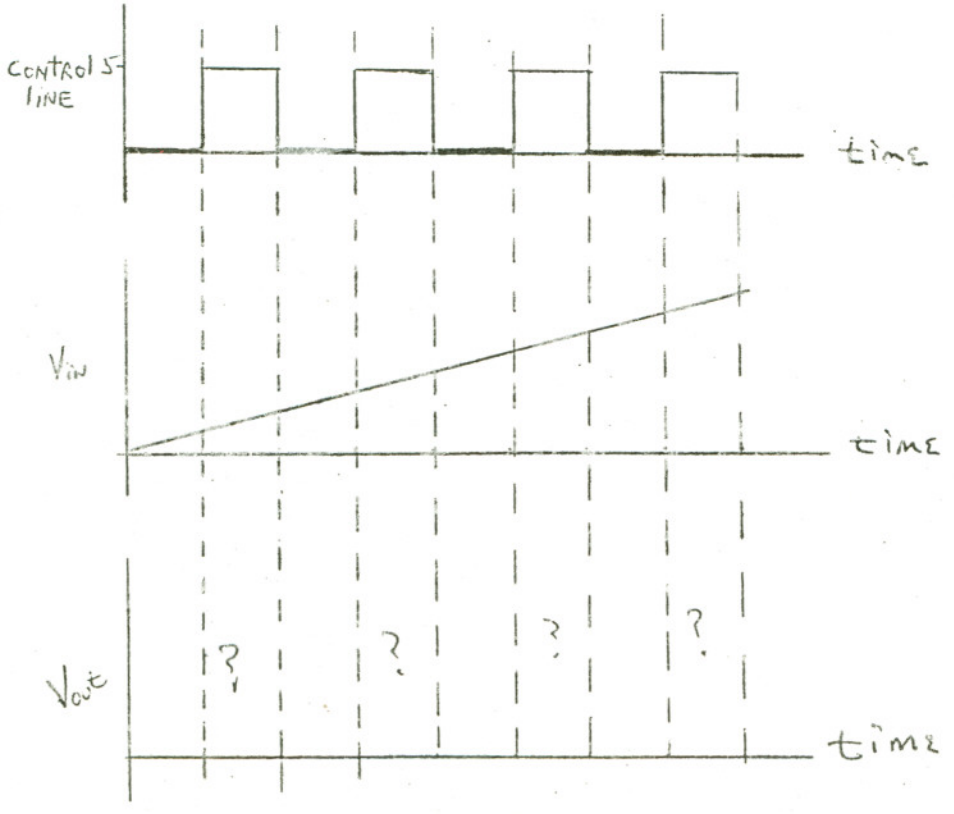

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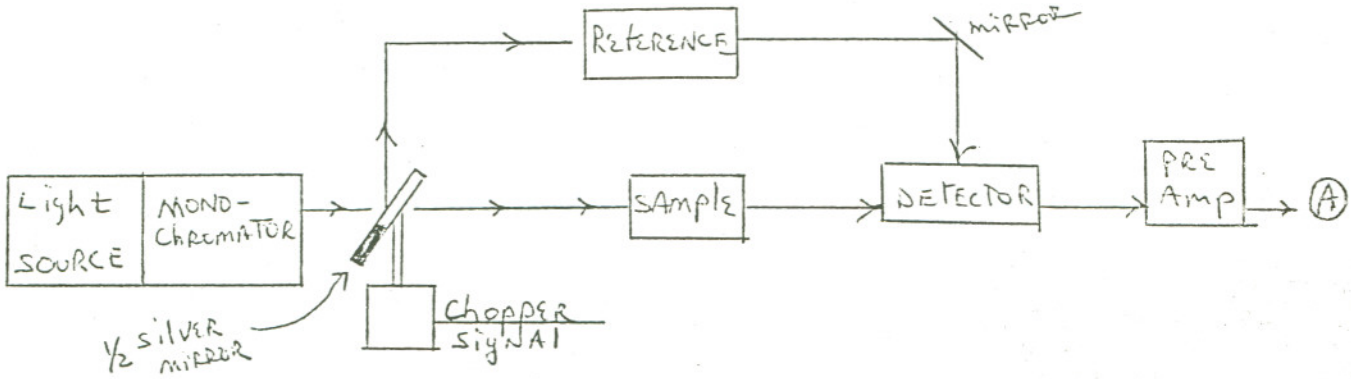
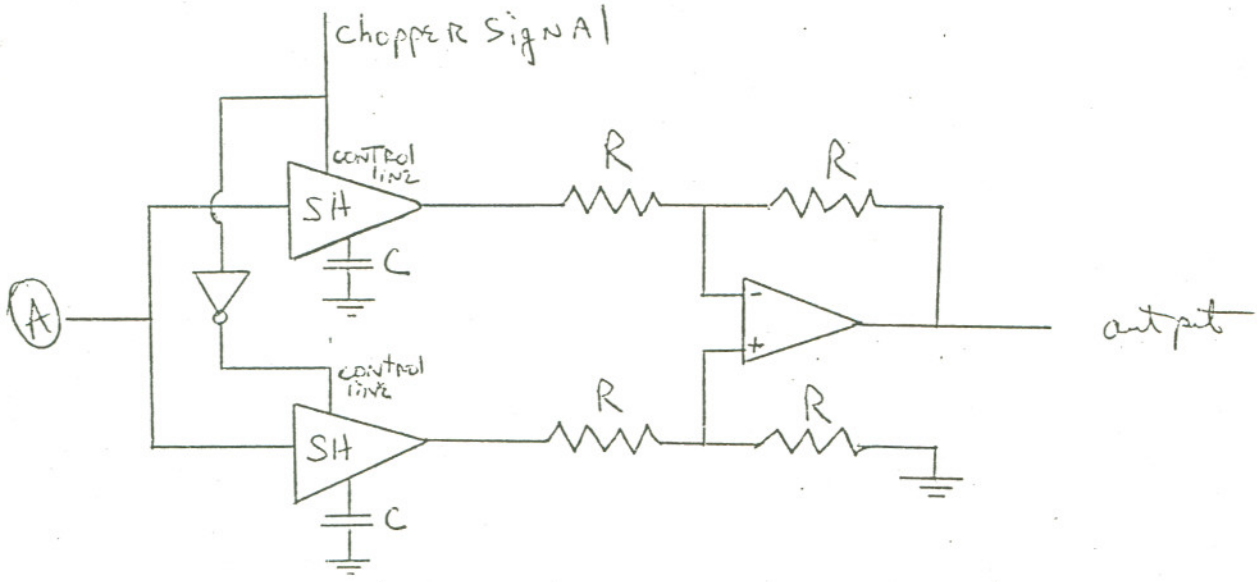
4) A simplified diagram for an inexpensive spectrometer is shown below.



The source light is time shared between the sample and reference cells. Light from both cells converge on a common detector. The resulting signal, proportional to optical absorbance, expressed as a voltage is the output at point A. This signal becomes the input to the circuit shown below.



The chopper signal presents a TTL high when the light path is from the reference cell, a low when the light path is from the sample cell. The chopper rate is about 1000 Hertz.

Single beam absorption measurements from sample and reference cells are shown on the following page:

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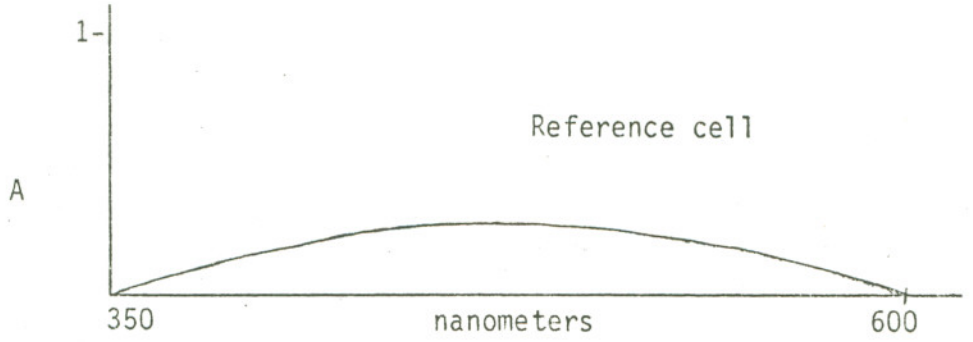

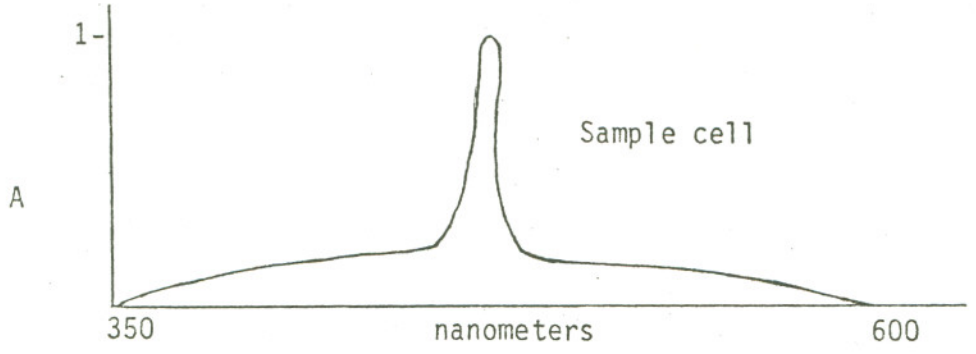

The time needed to complete a wavelength scan is several minutes.

a) Draw a plot showing output of the above circuit as the monochromator scans between 350 and 600 nanometers.

b) Explain the operation of this circuit using timing diagrams to support your answer.

NOTE: Questions 1, 2, 3, 4a, 4b are 20 points each.